

POKHARA UNIVERSITY

Level: Bachelor
Program: BE
Course: Calculus I

Semester: XXXX

Year : xxxx
Full Marks: 100
Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

MODEL QUESTION

1. a) If $y = a \cos(\log x) + b \sin(\log x)$ show that

$$x^2 y_{n+2} + (2n+1) x y_{n+1} + (n^2+1) y_n = 0. \quad [8]$$

- b) Assuming the validity of expansion, find the expansion of $e^{\sin x}$ using Maclaurin's series. [7]

OR

State Lagrange's mean value theorem and use it to prove:

$$\frac{b-a}{b} < \log\left(\frac{b}{a}\right) < \frac{b-a}{a}.$$

2. a) Define asymptotes. Find the asymptotes (if exists) of the curve $x^3 + 3x^2y - 4y^3 - x + y + 3 = 0$. [8]

- b) Find the radius of curvature at origin of the curve $4x^2 - 3xy + y^2 - 3y = 0$. [7]

3. Solve the following (Any Three) [15]

a) $\int \frac{dx}{1 - \cos x + \sin x}$

b) $\int_0^1 \cot^{-1}(1 - x + x^2) dx$

c) $\int_1^\infty \frac{\log x}{x^2} dx$

d) Evaluate by using gamma function $\int_0^{\frac{\pi}{6}} \cos^4 3\theta \sin^2 6\theta d\theta$.

4. a) Find the area of the surface of the solid generated by rotation of the curve $x = \cos t$, $y = 1 + \sin t$ for $0 \leq t \leq 2\pi$ about x-axis. [8]

- b) State Euler's theorem for two variables. For $u = \cos^{-1}\left(\frac{x+y}{\sqrt{x} + \sqrt{y}}\right)$,

show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + \frac{1}{2} \cot u = 0$. [7]

5. a) Find the extreme value of $f(x, y, z) = x^2 + y^2 + z^2$ satisfying the conditions $xy + yz + zx = 3a^2$. [8]

5. b) Solve the equation $\frac{dy}{dx} + \frac{y}{x}(\log y) = \frac{y}{x^2}(\log y)^2$. [7]

OR

If the temperature of a room is maintained at 10°C and the temperature of an object in the room falls from 100°C to 70°C in 20 minutes, find when the temperature of the object will be 30°C .

6. a) Find the solution of the initial value problem $6y'' - 5y' + y = 0$; $y(0) = 4$, $y'(0) = 0$. [7]

b) Solve the differential equation, $x^2y'' + xy' - 4y = x^2$. [8]

7. Attempt all [4×2.5 = 10]

a) Test the continuity of the function

$$f(x) = \begin{cases} x^2 - 2 & \text{for } x \leq 2 \\ x^2 - 4x + 6 & \text{for } x > 2 \end{cases} \text{ at } x = 2$$

b) Test the line of symmetry of the curve $x^3 + y^3 = 3axy$.

c) Solve the equation, $\sqrt{1-x^2} dy + \sqrt{1-y^2} dx = 0$.

d) Find the area included between the curves $x^2 = 4y$ and the line $x = 4y - 2$.